Skills Summary

Differentiating Accumulation Functions with the FTC

Use this reference while completing your worksheet or when studying.

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1. The basic FTC derivative (ftc-part2-basic)

If f is continuous and $g(x) = \displaystyle\int_a^x f(t)\,dt$ with a constant, then

$$g'(x) = f(x).$$

The integrand **is** the derivative — do not evaluate the integral first. Two variants to recognise:
$\displaystyle\int_x^b f(t)\,dt$ has derivative $-f(x)$ (flip the limits, pick up the minus sign), and $\displaystyle\int_a^x f(t)\,dt$ always satisfies $g(a) = 0$.

Example: $g(x) = \int_2^x (t^2+1)\,dt$. Find $g'(3)$.
Step 1. The lower limit is constant and the upper limit is x alone, so this is the plain FTC form and the derivative is just the integrand.
Step 2. $g'(x) = x^2 + 1$, so $g'(3) = 3^2 + 1 = \mathbf{10}$.

Try it: $g(x) = \int_5^x (2t-1)\,dt$. Find $g'(4)$.

check: **7**

2. The chain-rule form (ftc-chain)

If $g(x) = \displaystyle\int_a^{u(x)} f(t)\,dt$, then

$$g'(x) = f(u(x)) \cdot u'(x).$$

Substitute the whole upper limit into f, **then** multiply by its derivative. Missing the factor $u'(x)$ is the standard error — if the upper limit is anything other than a bare x, the factor is there.

Example: $g(x) = \int_0^{x^2} (t+1)\,dt$. Find $g'(3)$.
Step 1. The upper limit is $u(x) = x^2$, not x, so the chain-rule form applies: evaluate the integrand at $u(x)$ and multiply by $u'(x) = 2x$.
Step 2. $g'(x) = (x^2+1) \cdot 2x$, so $g'(3) = (9+1)(6) = \mathbf{60}$.

Try it: $g(x) = \int_1^{x^3} 2t\,dt$. Find $g'(2)$.

check: **192**

3. Evaluating accumulated change (accumulation-value)

When the question asks *how much* has accumulated (a volume, a distance, a total), you do evaluate — FTC Part 2:

$$\int_a^b f(t)\,dt = F(b) - F(a), \qquad F' = f.$$

Rate question → differentiate (Skill 1). Total question → integrate. Units follow the same rule: (units of f) × (units of t).

Example: Water enters at $r(t) = 4t + 2$ L/min. How much enters in the first 3 minutes?
Step 1. The question asks for a total, not a rate, so evaluate the accumulation $\int_0^3 r(t)\,dt$ rather than differentiating it.
Step 2. $F(t) = 2t^2 + 2t$, so the total is $F(3) - F(0) = (18 + 6) - 0 = \mathbf{24}$ litres.

Try it: $\int_0^4 (3t + 1)\,dt$ (litres, if $3t + 1$ is in L/min).

check: **28**

4. Reading g from f (accumulation-analysis)

Because $g' = f$, every question about the *shape* of g is a sign question about f:

$f > 0 \Rightarrow g$ increasing · $f < 0 \Rightarrow g$ decreasing · f changes $+$ to $-$ at $c \Rightarrow g$ has a maximum at c

f increasing $\Rightarrow g$ concave up. Justify with "$g' = f$ by the FTC", then argue from the sign of f — the sign change, not the zero, is what creates the extremum.

Example: $f(t) = t - 4$ and $g(x) = \int_0^x f(t)\,dt$ on $[0, 7]$. Where is g smallest?
Step 1. A minimum of g needs g' to change from negative to positive, and $g' = f$ by the FTC, so find where f changes sign.
Step 2. $f(t) = 0$ at $t = \mathbf{4}$; $f < 0$ before it and $f > 0$ after, so g decreases then increases and its minimum is at $x = 4$.

Try it: $f(t) = 6 - 2t$, $g(x) = \int_0^x f(t)\,dt$ on $[0, 5]$. Where is g largest?

check: $x = \mathbf{3}$

(!) Watch out: $g(x) = \int_a^x f(t)\,dt$ is a function of x only — t is a dummy variable and can never appear in the answer. And a variable *lower* limit flips the sign: $\frac{d}{dx}\int_x^b f = -f(x)$.